

Experiment 9

AIM: Implement the Private Ethereum Blockchain using Geth

THEORY:

Q1: What is Geth?

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Q2: Significance of a Private Ethereum Network

Setting up a Private Ethereum Network involves creating a customized blockchain environment

isolated from the public Ethereum network. It enables developers to test smart contracts,

consensus mechanisms, and network behavior without incurring gas costs or depending on

public validators.

Q3: Steps for creating a Private Ethereum Network

Steps Involved:

- **Choosing a Network ID:**

Each private blockchain must have a unique network ID to differentiate it from public or other private networks.

- **Choosing a Consensus Algorithm:**

Ethereum supports different consensus mechanisms such as Proof of Work (PoW) and Proof of Authority (PoA). In private networks, Clique (PoA) is often used for faster block confirmation.

- **Creating a Genesis Block:**

The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.

- **Initializing the Geth Database:**

The Geth init command initializes the blockchain using the genesis.json file, preparing the data directory for node operation.

- **Setting up Networking:**

Nodes communicate using enodes (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.

- **Running the Member Nodes:**

Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.

- **Running a Signer (in Clique):**

In the Clique consensus, designated signers validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

NOTE: Download the genesis file and edit the account details, including the public keys of all participating peers in the network.

PROCEDURE

Step 1: Install Geth

```
wget
```

```
https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
```

```
└─
```

```
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
```

```
mkdir -p ~/bin
```

```
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
```

```
export PATH="$HOME/bin:$PATH"
```

geth113 version

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use 'gcloud config set project [PROJECT_ID]'.
You can view your projects by running 'gcloud projects list'.
g2022_aditi_taksale@cloudshell:~$ wget
https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113

Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)[20.60.40.164]:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 28248613 (27M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz.1'

geth-linux-amd64-1.13.14-2bd6bd01.tar.gz.1 100%[=====>] 26.94M 4.86MB/s in 8.1s

2026-04-08 17:25:28 (3.31 MB/s) ~ 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz.1' saved [28248613/28248613]

geth-linux-amd64-1.13.14-2bd6bd01/
geth-linux-amd64-1.13.14-2bd6bd01/COPYING
geth-linux-amd64-1.13.14-2bd6bd01/geth
geth
Version: 1.13.14-stable
Git Commit: 2bd6bd01d2e8561dd7fc21b631f4a34ac16627a1
Git Commit Date: 20240227
Architecture: amd64
Go Version: go1.21.6
Operating System: linux
GOPATH=/home/g2022_ria_chaudhari/gopath/google/gopath
GOROOT=
```

Step 2: Create folders and accounts for both nodes

```
g2022_aditi_taksale@cloudshell:~$ mkdir -p ~/eth-private/node1 ~/eth-private/node2
g2022_aditi_taksale@cloudshell:~$ geth113 --datadir ~/eth-private/node1 account new
INFO [04-08|17:27:46.537] Maximum peer count ETH=50 total=50
INFO [04-08|17:27:46.540] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0xC8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878
Path of the secret key file: /home/g2022_ria_chaudhari/eth-private/node1/keystore/UTC--2026-04-08T17-27-52.865021965Z--c8fe5b129bb5ee23ec4bdb4cab363ed92a2f9878

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

```
g2022_aditi_taksale@cloudshell:~$ geth113 --datadir ~/eth-private/node2 account new
INFO [04-08|17:28:38.211] Maximum peer count ETH=50 total=50
INFO [04-08|17:28:38.214] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0x41911D883221851DD8C0Fe7C2a3E623D507bBb19
Path of the secret key file: /home/g2022_ria_chaudhari/eth-private/node2/keystore/UTC--2026-04-08T17-28-43.340511622Z--41911d883221851dd8c0fe7c2a3e623d507bbb19

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Step 3: Create password files

```
g2022_aditi_taksale@cloudshell:~$ echo "1234" > ~/eth-private/node1/password.txt
echo "1234" > ~/eth-private/node2/password.txt
```

Step 4: Create genesis.json and verify it looks correct

NODE1="PASTE_NODE1_ADDRESS_WITHOUT_0x_HERE"

NODE2="PASTE_NODE2_ADDRESS_WITHOUT_0x_HERE"

[illegible]

```
"alloc": {
  "0x${NODE1}": { "balance": "100000000000000000000" },
  "0x${NODE2}": { "balance": "100000000000000000000" }
}

EOF
```

```
g2022_aditi_taksale@cloudshell:~$ NODE1="C8Fe5b129bb5eE23eC4bDB4CaB363ed92a2f9878"
NODE2="41911d883221851DD8C0Fe7C2a3E623D507bBb1"
```

```
cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
```

[illegible]

Step 5: Initialize both nodes

```

g2022_aditi_taksale@cloudshell:~$ geth13 --datadir ~/eth-private/node1 init ~/eth-private/genesis.json
geth13 --datadir ~/eth-private/node2 init ~/eth-private/genesis.json
INFO [04-08]17:42:33.075] Maximum peer count ETH=50 total=50
INFO [04-08]17:42:33.079] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08]17:42:33.085] Set global gas cap cap=50,000,000
INFO [04-08]17:42:33.096] Initializing the RZG library backends=gokzg
INFO [04-08]17:42:33.219] Defaulting to pebble as the backing database
INFO [04-08]17:42:33.219] Allocated cache and file handles database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata cache=16.00MiB handles=16
INFO [04-08]17:42:33.300] Opened ancient database database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata/ancient/chain readonly=false
INFO [04-08]17:42:33.300] State schema set to default scheme=hash
INFO [04-08]17:42:33.301] Writing custom genesis block
INFO [04-08]17:42:33.304] Persisted trie from memory database nodes=4 size=478.00B time=3.369394ms gcnodes=0 gcsiz=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08]17:42:33.340] Successfully wrote genesis state database=chaindata hash=38dela..afc19d
INFO [04-08]17:42:33.340] Defaulting to pebble as the backing database
INFO [04-08]17:42:33.340] Allocated cache and file handles database=/home/g2022_ria_chaudhari/eth-private/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [04-08]17:42:33.400] Opened ancient database database=/home/g2022_ria_chaudhari/eth-private/node1/geth/lightchaindata/ancient/chain readonly=false
INFO [04-08]17:42:33.400] State schema set to default scheme=hash
INFO [04-08]17:42:33.400] Writing custom genesis block
INFO [04-08]17:42:33.404] Persisted trie from memory database nodes=4 size=478.00B time=3.273769ms gcnodes=0 gcsiz=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08]17:42:33.438] Successfully wrote genesis state database=lightchaindata hash=38dela..afc19d
INFO [04-08]17:42:33.515] Maximum peer count ETH=50 total=50
INFO [04-08]17:42:33.518] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08]17:42:33.523] Set global gas cap cap=50,000,000
INFO [04-08]17:42:33.524] Initializing the RZG library backends=gokzg
INFO [04-08]17:42:33.701] Defaulting to pebble as the backing database
INFO [04-08]17:42:33.701] Allocated cache and file handles database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata cache=16.00MiB handles=16
INFO [04-08]17:42:33.758] Opened ancient database database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata/ancient/chain readonly=false
INFO [04-08]17:42:33.758] State schema set to default scheme=hash
INFO [04-08]17:42:33.758] Writing custom genesis block
INFO [04-08]17:42:33.762] Persisted trie from memory database nodes=4 size=478.00B time=3.381802ms gcnodes=0 gcsiz=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08]17:42:33.798] Successfully wrote genesis state database=chaindata hash=38dela..afc19d
INFO [04-08]17:42:33.798] Defaulting to pebble as the backing database
INFO [04-08]17:42:33.798] Allocated cache and file handles database=/home/g2022_ria_chaudhari/eth-private/node2/geth/lightchaindata cache=16.00MiB handles=16
INFO [04-08]17:42:33.853] Opened ancient database database=/home/g2022_ria_chaudhari/eth-private/node2/geth/lightchaindata/ancient/chain readonly=false
INFO [04-08]17:42:33.853] State schema set to default scheme=hash
INFO [04-08]17:42:33.853] Writing custom genesis block
INFO [04-08]17:42:33.857] Persisted trie from memory database nodes=4 size=478.00B time=3.539219ms gcnodes=0 gcsiz=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-08]17:42:33.891] Successfully wrote genesis state database=lightchaindata hash=38dela..afc19d
g2022_aditi_taksale@cloudshell:~$

```

Step 6: Run Node 1 (signer)

```

g2022_aditi_taksale@cloudshell:~$ geth13 --datadir ~/eth-private/node1 \
--networkid 1234 \
--port 30303 \
--http --http.port 8545 \
--http.api personal,eth,net,web3,miner \
--unlock 0x$(NODE1) \
--password ~/eth-private/node1/password.txt \
--mine \
--miner.etherbase 0x$(NODE1) \
--allow-insecure-unlock \
console
INFO [04-08]17:44:52.561] Maximum peer count ETH=50 total=50
INFO [04-08]17:44:52.564] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08]17:44:52.568] Set global gas cap cap=50,000,000
INFO [04-08]17:44:52.569] Initializing the RZG library backends=gokzg
INFO [04-08]17:44:52.700] Allocated trie memory caches clean=154.00MiB dirty=256.00MiB
INFO [04-08]17:44:52.701] Using pebble as the backing database
INFO [04-08]17:44:52.701] Allocated cache and file handles database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [04-08]17:44:52.730] Opened ancient database database=/home/g2022_ria_chaudhari/eth-private/node1/geth/chaindata/ancient/chain readonly=false
INFO [04-08]17:44:52.730] State schema set to already existing scheme=hash
INFO [04-08]17:44:52.732] Initializing Ethereum protocol network=1234 dbversion=8
INFO [04-08]17:44:52.733] -----
INFO [04-08]17:44:52.733] Chain ID: 1234 (unknown)
INFO [04-08]17:44:52.733] Consensus: Clique (proof-of-authority)
INFO [04-08]17:44:52.733]
INFO [04-08]17:44:52.733] Pre-Merge hard forks (block based):
INFO [04-08]17:44:52.733] - Homestead: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [04-08]17:44:52.733] - Tangerine Whistle (EIP 150): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)
INFO [04-08]17:44:52.733] - Spurious Dragon/1 (EIP 155): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [04-08]17:44:52.733] - Spurious Dragon/2 (EIP 158): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [04-08]17:44:52.733] - Byzantium: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
INFO [04-08]17:44:52.733] - Constantinople: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/constantinople.md)
INFO [04-08]17:44:52.733] - Petersburg: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/petersburg.md)
INFO [04-08]17:44:52.733] - Istanbul: #nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/istanbul.md)
INFO [04-08]17:44:52.733] - Berlin: #nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/berlin.md)
INFO [04-08]17:44:52.733] - London: #nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/london.md)

```



```

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
coinbase: 0xc8fe5b129bb5ee23ec4bdb4cab363ed92a2f9878
at block: 1 (Wed Apr 08 2026 17:44:54 GMT+0000 (UTC))
datadir: /home/g2022_ria_chaudhari/eth-private/node1
modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-08|17:44:54.582] New local node record
INFO [04-08|17:44:59.007] Successfully sealed new block
INFO [04-08|17:44:59.008] Commit new sealing work
INFO [04-08|17:45:02.925] Looking for peers
INFO [04-08|17:45:04.007] Successfully sealed new block
INFO [04-08|17:45:04.008] Commit new sealing work
INFO [04-08|17:45:09.008] Successfully sealed new block
INFO [04-08|17:45:09.008] Commit new sealing work
INFO [04-08|17:45:12.929] Looking for peers
INFO [04-08|17:45:14.009] Successfully sealed new block
INFO [04-08|17:45:14.009] Commit new sealing work
INFO [04-08|17:45:19.008] Successfully sealed new block
INFO [04-08|17:45:19.009] Commit new sealing work
INFO [04-08|17:45:23.331] Looking for peers
INFO [04-08|17:45:24.008] Successfully sealed new block
INFO [04-08|17:45:24.009] Commit new sealing work
INFO [04-08|17:45:29.008] Successfully sealed new block
INFO [04-08|17:45:29.009] Commit new sealing work
INFO [04-08|17:45:33.555] Looking for peers
seq=1,775,670,194,395 id=ab0dd2ac241e15ee ip=34.21.175.36 udp=30303 tcp=30303
number=2 sealhash=163e73..f7277c hash=87483e..f8f261 elapsed=4.954s
number=3 sealhash=ce7107..eda36e txs=0 gas=0 fees=0 elapsed="395.747µs"
peercount=0 tried=108 static=0
number=3 sealhash=ce7107..eda36e hash=f92c48..6be3ea elapsed=4.999s
number=4 sealhash=346563..38e33e txs=0 gas=0 fees=0 elapsed="418.854µs"
number=4 sealhash=346563..38e33e hash=b736c7..b322b8 elapsed=4.999s
number=5 sealhash=4fa3a0..c18fe8 txs=0 gas=0 fees=0 elapsed="388.844µs"
peercount=0 tried=82 static=0
number=5 sealhash=4fa3a0..c18fe8 hash=2cd841..fe601a elapsed=5.000s
number=6 sealhash=a48aa5..fc52db txs=0 gas=0 fees=0 elapsed="336.108µs"
number=6 sealhash=a48aa5..fc52db hash=f436f2..354195 elapsed=4.999s
number=7 sealhash=02b9f2..f772ac txs=0 gas=0 fees=0 elapsed="497.656µs"
peercount=0 tried=26 static=0
number=7 sealhash=02b9f2..f772ac hash=5ca646..bfc786 elapsed=4.999s
number=8 sealhash=212487..bcd33f txs=0 gas=0 fees=0 elapsed="366.968µs"
number=8 sealhash=212487..bcd33f hash=b48c05..084ec5 elapsed=4.999s
number=9 sealhash=b6c8db..a8a196 txs=0 gas=0 fees=0 elapsed="337.61µs"
peercount=0 tried=91 static=0

```

Step 7: Open new terminal tab and run Node 2 (member)

```

g2022_aditi_taksale@cloudshell:ch13 --datadir ~/eth-private/node2 \
--noWorkid 1234 \
--port 30304 \
--http --http.port 8546 \
--authrpc.port 8552 \
--http.api personal,eth,net,web3 \
--unlock 0x5(NODE2) \
--password ~/eth-private/node2/password.txt \
--allow-insecure-unlock \
console
INFO [04-08|17:48:52.981] Maximum peer count
INFO [04-08|17:48:52.985] Smartcard socket not found, disabling
INFO [04-08|17:48:52.989] Set global gas cap
INFO [04-08|17:48:52.989] Initializing the KEG library
INFO [04-08|17:48:53.126] Allocated trie memory caches
INFO [04-08|17:48:53.126] Using ptable as the backing database
INFO [04-08|17:48:53.127] Allocated cache and file handles
INFO [04-08|17:48:53.151] Opened ancient database
INFO [04-08|17:48:53.151] State scheme set to already existing
INFO [04-08|17:48:53.153] Initialising Ethereum protocol
INFO [04-08|17:48:53.154]
INFO [04-08|17:48:53.154] Chain ID: 1234 (unknown)
INFO [04-08|17:48:53.154] Consensus: Clique (proof-of-authority)
INFO [04-08|17:48:53.154]
INFO [04-08|17:48:53.154] Pre-Merge hard forks (block based):
INFO [04-08|17:48:53.154] - Homestead: #0
INFO [04-08|17:48:53.154] - Tangerine Whistle (EIP 150): #0
INFO [04-08|17:48:53.154] - Spurious Dragon/1 (EIP 155): #0
INFO [04-08|17:48:53.154] - Spurious Dragon/2 (EIP 158): #0
INFO [04-08|17:48:53.154] - Byzantium: #0
INFO [04-08|17:48:53.154] - Constantinople: #0
INFO [04-08|17:48:53.154] - Petersburg: #0
INFO [04-08|17:48:53.154] - Istanbul: #nil>
INFO [04-08|17:48:53.154] - Berlin: #nil>
INFO [04-08|17:48:53.154] - London: #nil>
INFO [04-08|17:48:53.154]
INFO [04-08|17:48:53.154] The Merge is not yet available for this network!
ETH=50 total=50
err="stat /run/pcscd/pcscd.comm: no such file or directory"
cap=50,000,000
backend=gokzg
clean=154.00MiB dirty=256.00MiB
database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata cache=512.00MiB handles=524,288
database=/home/g2022_ria_chaudhari/eth-private/node2/geth/chaindata/ancient/chain readonly=false
scheme=hash
network=1234 dversion=8
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/constantinople.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/petersburg.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/istanbul.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/berlin.md)
(https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/london.md)

```

```

68034ec63cf3238e127.0.0.1:30304
INFO [04-08|17:48:53.244] WebSocket enabled
INFO [04-08|17:48:53.244] HTTP server started
INFO [04-08|17:48:54.532] Unlocked account
WARN [04-08|17:48:54.584] Served eth_coinbase
Welcome to the Geth JavaScript console!
instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
at block: 0 (Thu Jan 01 1970 00:00:00 GMT+0000 (UTC))
datadir: /home/g2022_ria_chaudhari/eth-private/node2
modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-08|17:48:55.861] New local node record
INFO [04-08|17:49:03.289] Looking for peers
INFO [04-08|17:49:13.303] Looking for peers
INFO [04-08|17:49:24.592] Looking for peers
INFO [04-08|17:49:34.781] Looking for peers
url=ws://127.0.0.1:8552
endpoint=127.0.0.1:8552 auth=true prefix=core=localhost vhost=localhost
address=0x41911d883221851D08C0Fe7C2a3E623D507Bb19
reqid=3 duration="52.151µs" err="etherbase must be explicitly specified"
seq=1,775,670,478,928 id=3c0cea706c0e875d ip=34.21.175.36 udp=30304 tcp=30304
peercount=1 tried=60 static=0
peercount=0 tried=75 static=0
peercount=0 tried=39 static=0
peercount=0 tried=67 static=0

```

Step 8: Connect the two nodes In Node 1 console, get its enode

```

> admin.nodeInfo.enode
"enode://37a3db1c1e34eeb3611941ca5b7001168ac8ee28506649e6a5290979c7a78e7277565b37ec5d703871a9970a4794c51bde075d7991723256e0c07c8cb367eb834.21.175.36:30303"

```

Now, switch to Node 2 console and paste the Node 1 enode

```

> admin.addPeer("enode://37a3db1c1e34eeb3611941ca5b7001168ac8ee28506649e6a5290979c7a78e7277565b37ec5d703871a9970a4794c51bde075d7991723256e0c07c8cb367eb834.21.175.36:30303")
true

```

In Node 2 console, check they are connected

```
> admin.peersINFO [04-08|18:01:18.017] imported new chain segment      number=196 hash=9ac569..83e85f blocks=1   txs=0  mgas=0.000 elapsed=7.74ms  mgasps=0.000 trieditt
[
  {
    caps: ["eth/65", "snap/1"],
    enode: "enode://37a3db1c1e34eac3611941ca5b7001168ac8eac28506649e6a5290979c7a78e7277565b37ec5d703871a9970a4794c51bde075d7991723256e0c07c8cb367eb8127.0.0.1:30303",
    id: "ab0dd2ac241e15eac929ff71bbb9d19f854213e77207f9ee0b74c4911801aef7",
    name: "geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6",
    network: {
      inbound: false,
      localAddress: "127.0.0.1:60510",
      remoteAddress: "127.0.0.1:30303",
      static: true,
      trusted: false
    },
    protocols: {
      eth: {
        version: 65
      },
      snap: {
        version: 1
      }
    }
  }
]
```

Step 9: Send a transaction In Node 1 console, send transaction to Node 2

```
> eth.sendTransaction({
.....  from: "0xc8Fe5b129bb5e23ec4bD84CaB363ed92a2f9878",
.....  to: "0x41911D883221851DD8C0Fe7C2a3E623D507bBb19",
.....  value: "0xde0b6b3a7640000"
..... })
INFO [04-08|18:03:16.769] Setting new local account      address=0xc8Fe5b129bb5e23ec4bD84CaB363ed92a2f9878
INFO [04-08|18:03:16.770] Submitted transaction      hash=0x6e201621cef1dc5e3c3f114cb03055c23b1f083ebf98411f8c91cb48e47265 from=0xc8Fe5b129bb5e23ec4bD84CaB363ed92a2f9878 nonce=
0 recipient=0x41911D883221851DD8C0Fe7C2a3E623D507bBb19 value=1,000,000,000,000,000
"0x6e201621cef1dc5e3c3f114cb03055c23b1f083ebf98411f8c91cb48e47265"
```

and check the balance of Node 2

```
> web3.fromWei(
...   eth.getBalance("0x41911D883221851DD8C0Fe7C2a3E623D507bBb19"),
...   "ether"
... )
101
```

CONCLUSION:

Through this experiment, two smart contracts - OrderRecord and ReviewVerification, were successfully created and deployed on the Ethereum Sepolia test network using Solidity and Remix IDE. The experiment demonstrated how blockchain technology can be integrated into a full-stack web application to permanently and transparently record order transactions and product reviews. The use of the Sepolia test network and MetaMask allowed safe testing without real funds. Overall, this experiment provided practical understanding of designing, compiling, deploying, and testing smart contracts on the Ethereum blockchain.